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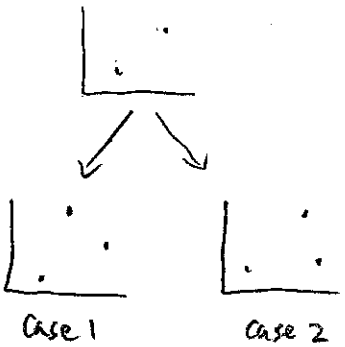
HW#4

Problem 1a

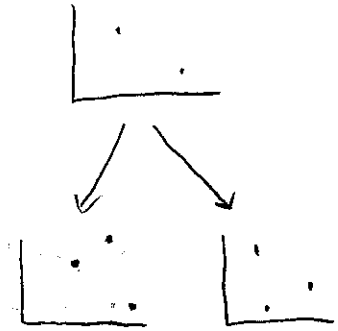
Suppose f is not monotonic and continuous. There are 4 cases for f not being monotonic.

let $a, b, c \in X$ with $a < b < c$.

If f is not strictly increasing; if f is not strictly decreasing.



Case 1 Case 2



Case 3 Case 4.

We want to prove in case 3, and the argument works for other cases.

Suppose $f(c) < f(a) < f(b)$. We want to show if f is not monotonic, then $f((a,c))$ is not open. So our goal is to find a global maximum in (a, c). Take $\max\{f([a,c])\}$. Since $f(c) < f(a) < f(b)$, clearly, $f(a)$ and $f(c)$ are not max, so there exist a point $m \in (a,c)$ such that $f(m)$ is max in $f((a,c))$. This implies $f(m)$ is not an interior point of $f((a,c))$ becausse $\exists \delta > 0$ such that $(f(m) - \delta, f(m) + \delta) \notin f((a,c))$. Thus, we can conclude that $f((a,c))$ is not open.

Problem 1b

We want to show that $\overline{f(E)} = f(X)$ by defination of dense.

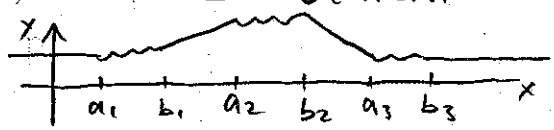
Take $x_n \to x$ with $x_n \in E$ and $x \in X - E$. Then by continuity, $f(x_n) \to f(x)$. Since $f(x_n) \in f(E)$ and $f(x) \in f(X)$, this implies that $f(E)$ and its limit points are in $f(X)$. So $\overline{f(E)} = f(X)$. Therefore, $f(E)$ is dense in $f(X)$.

(c) To prove $f=g$ on all of X, we only need to show that $f=g$ for $p \in X \setminus E$. Since E is dense in X, there exist sequence $\{q_n\}$ in E such that $\lim_{n\to\infty} q_n = p$ and $q_n \neq p$ for all n. Because p is a limit point for E implies every neighborhood $B_\varepsilon(p)$ contains a point $q \neq p$ such that $q \in E$. Hence, we can get that:

$$f(p) = f(\lim_{n\to\infty} q_n) = \lim_{n\to\infty} f(q_n) = \lim_{n\to\infty} g(q_n) \quad \text{since we are given } f=g \text{ for } x \in E.$$
$$= g(\lim_{n\to\infty} q_n)$$
$$= g(p)$$

Therefore, $f(p) = g(p)$ for all $p \in X$.

(d) Consider $E^c = \bigcup (a_i, b_i)$, where $i \in \mathbb{Z}$ and $a_i < b_i < a_{i+1} < b_{i+1} \cdots$



Define $\tilde{f}(x) = \begin{cases} f(x), & x \in E \\ f(b_i) + (x-b_i)\dfrac{f(a_{i+1}) - f(b_i)}{a_{i+1} - b_i}, & x \in E^c \end{cases}$

We need to show $\tilde{f}$ is continuous at a_i and b_i for $\forall i$.

We first show that $\tilde{f}$ is continuous at b_i. For $x < b_i$, $\tilde{f}(x) = f(x)$. Since $f(x)$ is continuous, so $\tilde{f}(x)$ is continuous. For $x > b_i$, take $\varepsilon > 0$, we need to find some δ such that $|x - b_i| < \delta$ implies $|\tilde{f}(x) - f(b_i)| < \varepsilon$. Then we get that $\left| (x-b_i) \dfrac{f(a_{i+1}) - f(b_i)}{a_{i+1} - b_i} \right| < \varepsilon$. Since $|x - b_i| < \delta$, so $\delta \left| \dfrac{f(a_{i+1}) - f(b_i)}{a_{i+1} - b_i} \right| < \varepsilon$

$\delta < \varepsilon \cdot \left| \dfrac{a_{i+1} - b_i}{f(a_{i+1}) - f(b_i)} \right|$. So $\tilde{f}(x)$ is continuous at b_i. Secondly, we need to show that $\tilde{f}(x)$ is continuous at a_i. For $x > a_i$, $\tilde{f}(x) = f(x)$, so it's continuous. For $x < a_i$, take $\varepsilon > 0$, we need to find some δ such that $|x - a_i| < \delta$ implies $|\tilde{f}(x) - f(a_i)| < \varepsilon$. Then we get $\left| f(b_i) - f(a_i) + (x - b_i) \dfrac{f(a_{i+1}) - f(b_i)}{a_{i+1} - b_i} \right|$ as $|x - a_i| < \delta$. We can find such δ, so $\tilde{f}(x)$ is continuous at a_i.

Therefore, $\tilde{f}(x)$ is continuous at a_i and b_i for $\forall i$.

If E is not closed, then $\tilde{f}$ may not have continuous extension. For example, take $E = (0,1)$ and $f(x) = \frac{1}{x}$ on E. Then $\tilde{f}(x) = f(x)$ for $x \in E$. However, $\tilde{f}(x)$ is not continuous at $x = 0$ since $\tilde{f}(x) \to \infty$ as $x \to 0$.

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1(d) $E \subseteq \mathbb{R}$ is a closed subset, $f: E \to \mathbb{R}$ is continuous. Prove $\exists \tilde{f}: \mathbb{R} \to \mathbb{R}$ s.t. $\tilde{f}|_E = f$.

Since E is closed, by Prop. 3.17 E^c is open. By Prop 3.18, E^c can be written as a union of disjoint open intervals, say $E^c = \bigcup (a_n, b_n)$, where $a_n, b_n \in E$ (the "endpoints" are in E since the intervals are open and disjoint).

Define $\tilde{f}(x)$ as

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in E \\ \left(\frac{x-a_n}{b_n-a_n}\right) f(b_n) + \left(\frac{b_n-x}{b_n-a_n}\right) f(a_n) & \text{if } x \in [a_n, b_n] \end{cases}$$

Now $\tilde{f}(x)$ is well-defined and $\forall x \in (a_n, b_n)$, either $f(a_n) \le \tilde{f}(x) \le f(b_n)$ or $f(b_n) \le \tilde{f}(x) \le f(a_n)$ depending on whether $f(a_n) \le f(b_n)$ or $f(b_n) \le f(a_n)$.

Now suppose $x \in E$ and fix $\varepsilon > 0$. There are 2 cases to show $\tilde{f}$ is continuous.

Case 1: Suppose that for every $M > 0$, $(x-M, x) \cap E \neq \emptyset$ and $(x, x+M) \cap E \neq \emptyset$.

Now since $\tilde{f}|_E = f$, $\exists \delta > 0$ s.t. if $y \in E$ and $|x-y| < \delta$ then $|\tilde{f}(x) - \tilde{f}(y)| < \varepsilon$.

Because of the condition for case 1, we may choose some $x_1, x_2 \in E$ with $x - \delta < x_1 < x < x_2 < x + \delta$. Choose $\delta' = \min\{x - x_1, x_2 - x\}$. If $|y - x| < \delta'$, then if $y \in E$ we're done. If $y \in E^c$, then $y \in (a_m, b_m)$ for some $m \in \mathbb{N}$. Furthermore, $a_m, b_m \in E$ and are within δ of x. Also, $\tilde{f}(y)$ is between $f(a_m)$ and $f(b_m)$. Thus $\tilde{f}(y)$ is within ε of $\tilde{f}(x)$ since $\tilde{f}(a_m) = f(a_m)$ and $\tilde{f}(b_m) = f(b_m)$ are within ε of $\tilde{f}(x)$.

Case 2: Suppose there exists some $M > 0$ for which $(x-M, x) \cap E = \emptyset$ and $(x, x+M) \cap E = \emptyset$. In this case, x is the endpoint of one of the intervals of E^c. Thus $\tilde{f}$ is linear on either $[x, x+M)$ or $(x-M, x]$. We can then find a $\delta > 0$ corresponding to ε on this side of x. For the other side, take the argument from Case 1 to get some δ'. Then choose $\delta'' = \min\{\delta, \delta'\}$. This proves $\tilde{f}(x)$ is continuous $\forall x \in \mathbb{R}$.

If E is not closed, f might not have a continuous extension. Take $E = (0,1)$ and $f(x) = \frac{1}{x}$ on $(0,1)$. If f has a continuous extension $\tilde{f}$ then $f(x) = \tilde{f}(x)$ for $x \in E$. Taking $\lim_{x \to 0^+} x$, we know $\tilde{f}(0^+)$ does not exist. Hence any extension of f cannot be continuous at 0.

1(e) Notice that the sets $\overline{f(B_{\frac{1}{n}}(p))} \supseteq \overline{f(B_{\frac{1}{n+1}}(p))} \ldots$ form a nested sequence of closed sets. Fix $\varepsilon > 0$. Since f is uniformly continuous, $\exists \delta > 0$ s.t. $d(p,q) < \delta \Rightarrow d(f(p), f(q)) < \varepsilon$ for $p, q \in E$. Take n large enough so that $\frac{2}{n} < \delta$, then for $q, r \in B_{\frac{1}{n}}(p)$,

$$d(q,r) \le d(q,p) + d(p,r) < \frac{2}{n} < \delta$$

so $d(f(q), f(r)) < \varepsilon$. Thus $f(B_{\frac{1}{n}}(p))$ is bounded in $\mathbb{R}$, so $\overline{f(B_{\frac{1}{n}}(p))}$ is bounded as well. Therefore, we can see that for large enough n the sets form a compact nested sequence. Since $B_{\frac{1}{n}}(p)$ has diameter $\frac{2}{n}$, taking $n > \frac{2}{\delta}$ would imply

$$\operatorname{diam} f(B_{\frac{1}{n}}(p)) = \operatorname{diam} \overline{f(B_{\frac{1}{n}}(p))} < \varepsilon$$

(Prop. 7.17) So $\lim_{n \to \infty} \operatorname{diam} \overline{f(B_{\frac{1}{n}}(p))} = 0$ and thus $\bigcap_n f(B_{\frac{1}{n}}(p))$ consists of a single point. Also since $f(B_{\frac{1}{n}}(p)) \to 0$ as $n \to \infty$, we see that by choosing points arbitrarily close to p, their images under $\tilde{f}$ are arbitrarily close to $\tilde{f}(p)$. Explicitly, let δ be small enough s.t. for $x, y \in E$ then $d(x,y) < 2\delta$ implies $d(f(x), f(y)) < \varepsilon/3$. Choose n large enough that $\frac{1}{n} < \delta$ and thus for any $x, y \in B_{\frac{1}{n}}(p)$, $d(x,y) < 2/n < 2\delta$ so $\operatorname{diam} f(B_{\frac{1}{n}}(p)) < 2\varepsilon/3$, so $d(f(x), \tilde{f}(p)) < 2\varepsilon/3$, for any p.

To show $\tilde{f}$ is uniformly continuous, fix $\varepsilon > 0$. Since f is uniformly continuous, $\exists \delta > 0$ such that $d(r,s) < \delta$ implies $d(f(r), f(s)) < \varepsilon/3$. Now let $p, q \in X$ be any points such that $d(p,q) < \delta/3$. Choose n large enough so that $n > \frac{3}{\delta}$, and both $d(\tilde{f}(r), \tilde{f}(p)) < \varepsilon/3$ and $d(\tilde{f}(s), \tilde{f}(q)) < \varepsilon/3$ for $r \in B_{\frac{1}{n}}(p)$ and $s \in B_{\frac{1}{n}}(q)$.

Also,

$$d(r,s) \le d(r,p) + d(p,q) + d(q,s) < \delta$$

so $d(f(r), f(s)) = d(\tilde{f}(r), \tilde{f}(s)) < \varepsilon/3$. By the triangle inequality, $d(\tilde{f}(p), \tilde{f}(q)) < \varepsilon$ so $\tilde{f}$ is uniformly continuous and thus continuous on X, and $\tilde{f}|_E = f$.

Problem 1e,f

(e) Another solution:

Since f is uniformly continuous, if x_n is Cauchy then $f(x_n)$ is Cauchy.

Consider: $f: E \to Y$, $E \subseteq X$, f uniformly continuous.

① Want to define $\tilde{f}(x)$ for $x \in X - E$. Take a sequence $x_n \to x$. This is Cauchy. If Y is <u>complete</u> then $f(x_n)$ is Cauchy and converges, say to p.

Need to show that this is not dependent on choice of sequence. Say $y_n \to x$ and $f(y_n) \to q$.

For $\varepsilon > 0$, want to show $d(p,q) = 0$. Fix $\varepsilon > 0$.

$$d(p,q) \leq d(p, f(x_n)) + d(f(x_n), f(y_n)) + d(f(y_n), q).$$

For $n > N$ large, $d(p, x_n) < \varepsilon$
$d(y_n, q) < \varepsilon$

Also, by uniform continuity, there is $\delta > 0$ such that $d(f(x_n), f(y_n)) < \varepsilon$ for $d(x_n, y_n) < \delta$.

Since $x_n \to x$, $y_n \to x$, for large n, $d(x_n, y_n) < d(x_n, x) + d(x, y_n) < \delta$.
$d(x_n, x) < \frac{\delta}{2}$
$d(y_n, x) < \delta/2$

So, choosing large enough N, one has that $d(p,q) < 3\varepsilon$.

This is true for all ε, so $d(p,q)=0$.

Thus, $\tilde{f}(x)$ is well-defined. It is continuous since any sequence converging to x, say x_n, has $f(x_n)$ converging to $f(x)$, as we have just shown.

(f) One can replace $\mathbb{R}$ with any complete metric space Y.

If Y is not complete, may not be true.

e.g.: $X=\mathbb{R}$ $Y=\mathbb{Q}$
$E=\mathbb{Q}$.

$f:\mathbb{Q}\longrightarrow\mathbb{Q}$ identity map

$\tilde{f}:\mathbb{R}\longrightarrow\mathbb{Q}$ cannot extend continuously.

~~Lemma: Any continuous map $g:\mathbb{R}\to\mathbb{Q}$ must be constant.~~